

Overview of the Current State of Dark Matter Research

Oliver Townsend
Physics and Astronomy
University of Southampton

January 2026

Abstract

Dark matter is a very widely dispersed physics problem, bringing in many people from many different disciplines, all with separate theories and experiments. As a result, it is hard to get an idea about the overall current state of dark matter research. In this report I will quickly summarise the history of dark matter, then discuss some of the more popular candidate solutions, and end by using some current dark matter detectors as examples to explain different dark matter related experimental techniques.

1 Introduction

2 A Very Brief History of Dark Matter

2.1 Early Development

Before we dive into the modern state of dark matter research, I'll catch you up to speed, on the history of research up to this point. The phrase 'dark matter' might conjure up the idea of clouds of invisible particles in deep space, but for a long time it was very different. In the late 19th century and a lot of the 20th century, the phrase dark matter was used to refer to any matter in galaxies and galaxy clusters that might not be visible to the methods used at the time, such as white dwarfs, or clouds of gas. [1]

Many authors label Fritz Zwicky as the first to properly discover dark matter in the 1930s. Though he undoubtedly progressed the subject and introduced new ideas, the idea of dark matter was already in use before him, and his usage of the term does not really align with ours today. Zwicky noticed that the velocities of galaxies within the Coma cluster were widely dispersed, and used theories from thermodynamics, the virial theorem, to show that the expected mass of the cluster was far larger than the mass given by the summation of the masses of the individual galaxies, his reference to this mass as dark matter aligned with thought at the time about invisible sources of matter in space, and served as an introduction of the use of thermodynamic methods in the modeling of galaxies. [1]

2.2 Galactic Rotation Curves

Work on invisible sources of mass continued throughout this time, but the next large progression in the field came when radio observations of galaxies became precise enough to measure the rotation curves of galaxies, famously by Vera Rubin, where outermost mass was moving far faster than expected, leading to the conclusion that there must be a significant amount of mass in the outermost section of a galaxy, at the time called a corona, halo, or massive envelope. By the 1980s, physicists agreed that there was some mechanism that caused the effect of extra mass outside the normal visible baryonic mass of galaxies and galaxy clusters. [1]

2.3 Modern Restrictions on Dark Matter

In the mid-1980s, through the novel use of computer simulation, hot dark matter, that is, dark matter candidate particles which must be relativistic when created in the early universe such as neutrinos (a promising candidate up to this point), were ruled out as a possible solution to the dark matter problem. It's configuration leads to larger galactic structures forming during development of the universe. Following searches for dark matter focused on warm or cold, meaning non-relativistic speed, dark matter, necessary for the formation of the structure of the universe we observe. [2] [3]

The 90s and 00s brought an increased quality of data of the Cosmic Microwave Background, giving us information about the structure of the early universe. This allowed great developments in the field of big bang nucleosynthesis, the study of the creation of nuclei in the early universe,

including the prediction of the baryon density of the universe, found to be too low to allow for baryonic dark matter. There are exceptions to this rule, such as black holes that formed prior to nucleosynthesis, but generally this rules out many early ideas about dark matter, such as dark clouds of gas, dim stars, or comets. [1]

Also influencing the study of dark matter is the very influential discovery of the bullet cluster in 2006. [4] The image of the bullet cluster in Figure 1 is an extremely convincing and clear representation of dark matter, showing the majority of mass in the galaxy system (using micro gravitational lensing) as being outside of the X-Ray emitting gas, the majority of the baryonic matter of the system. Since then, other galaxy collisions have also been found with this property [5]. Any theory of dark matter must explain these phenomena.

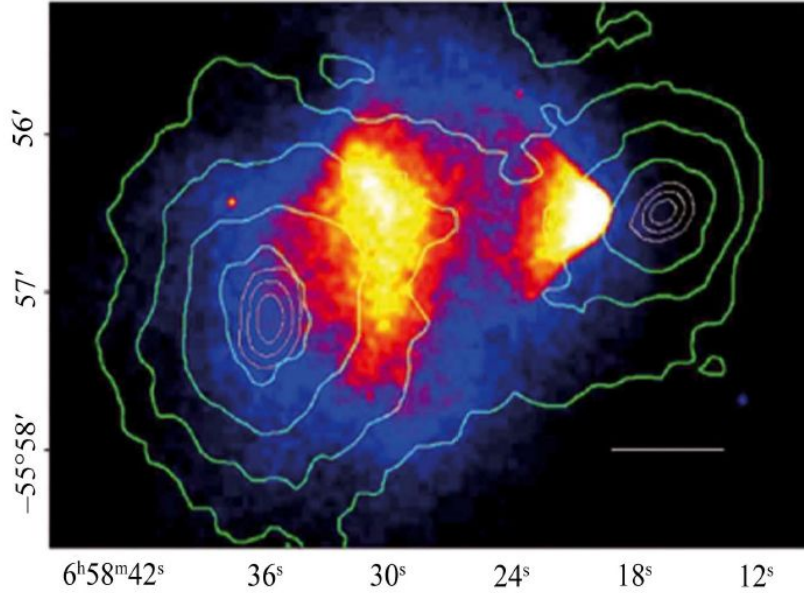


Figure 1: The bullet cluster. The coloured background shows the X-Ray emission from the two colliding galaxies, the contour lines show the weak gravitational lensing displaying the mass of the system. Pertinently, the x-Ray emission, from the baryonic matter of the galaxies, does not align with the majority of the baryonic matter of the system. [4]

These findings represent attempts to whittle down the possible options for dark matter as much as possible. A dark matter particle must be non-relativistic in the early universe (cold or warm), non-baryonic, interact only

weakly with regular matter, be long lived, and have self interaction. I will now give a quick overview of the current best candidates for dark matter, before moving onto efforts at detecting these particles.

3 The Candidates

3.1 WIMPs

The current most studied dark matter candidate is the WIMP (Weakly Interacting Massive Particles). This covers a range of particles outside the standard model, which interact through gravity and the weak nuclear force, or another force just as weak. What makes WIMPs so attractive as dark matter is that, in order to match the observed dark matter density of the universe, dark matter particles must self-annihilate with a cross section very similar to what arises naturally from interaction through the weak force. This coincidence, sometimes called the WIMP miracle, has motivated lots of research into these particles over a wide mass range. [3]

3.2 Supersymmetry

Dark matter candidate particles are often seen as more plausible if they arise from other areas of study, and happen to also explain the dark matter problem as a bonus. An example of this is one of the more popular WIMP candidates, the neutralino, the superpartner of the neutrino in supersymmetry (SUSY). Supersymmetry is an extension of the standard model which was originally developed to solve the gauge hierarchy problem of the standard model, and it predicts a corresponding superpartner for every boson and fermion. Neutrinos were considered a promising candidate for particle dark matter, but their light masses meant that they would be relativistic in the early universe, the lightest neutralino (the only stable neutralino), however, is heavier enough to be able to constitute cold dark matter [3]. Supersymmetry, like WIMPs generally, has started to fall out of fashion with the failures to detect any superpartners at the Large Hadron Collider, but some hope still prevails that supersymmetry can still solve the problem of dark matter. [6]

3.3 Axions

There are other successful theories in physics which provide possible dark matter candidate particles while explaining other areas, an important one is quantum chromodynamics (QCD) which predicts a particle called an axion. QCD is a highly successful theory which very well predicts the behaviour of the strong force. QCD predicts the violation of charge parity proportional to an angle θ , experiments on the electric dipole moment of the neutron restrict θ to smaller than 10^{-10} . Since θ comes from a sum of two seemingly unrelated terms, there must be some mechanism causing this cancellation, this problem is known as the strong CP problem. The most common solution involves introducing a new symmetry which drives the angle θ towards zero. The consequence of this very minimal solution is the existence of a Nambu-Goldstone boson, the axion. [3]

In order to be consistent with lab and cosmological observations, the axion must be lighter than 10^{-3} eV. Conveniently, this also corresponds to the approximate mass needed for axions to be a good dark matter candidate, allowing them to be stable over very long times, and allows for a production mechanism in the early universe, called misalignment production, which predicts the correct density of axions to constitute dark matter. Therefore, axions of a mass around 10^{-5} eV produced by misalignment production in the early universe are a very popular possible candidate particle. [1]

3.4 MOND

An alternative solution to the dark matter problem comes in the form of a modified theory of gravity to explain the seemingly excess mass in galaxies and galaxy clusters, often called Modified Newtonian Dynamics (MOND). One benefit of MOND over particle dark matter theories, and likely its most important prediction, is that it explains the Tully-Fisher relation, a regularity in the rotation curves of galaxies. [1]

MOND makes the prediction that below a certain critical acceleration a_* , Newtons law $F = ma$ becomes $F = ma^2/a_*$. Using Newtons equation for gravity, rearranging for acceleration, then substituting to get velocity, you find that

$$v(r) = (G\mathcal{M}_b(r)a_*)^{1/4} \quad (1)$$

where $\mathcal{M}_b(r)$ is the baryonic mass inside a sphere of radius r . This implies a relation between the baryonic mass of a galaxy and the rotational velocity

far from the galaxy of $v_\infty \propto \mathcal{M}^{1/4}$, a prediction which holds well for almost all galaxies. [3]

Particle dark matter can explain this law if there must be a set proportional relationship between the amount of dark matter and baryonic mass in each galaxy, explained through the process of structure formation, since already a larger dark matter halo would gravitationally attract a larger quantity of baryons. The current version of MOND is far more advanced, Tensor-Vector-Scalar gravity (TeVeS), unlike the original MOND, preserves conservation laws, and includes relativity, allowing it to be made consistent with gravitational lensing and other considerations. [1]

Despite these successes, MOND still faces problems. The bullet cluster, Figure 1 from earlier, shows the mass of the galaxies removed from the baryonic mass, something which particle dark matter can explain easily [4]. MOND also cannot explain all of the missing mass in galaxy clusters, and most modified gravity theories include some particle dark matter in order to fully explain these observations [7]. Finally, many galaxies have been found with no (very little) dark matter, explainable in particle theories, but harder to explain using MOND [8].

3.5 Primordial Black Holes

As mentioned earlier, although black holes technically constitute baryonic matter, primordial black holes that formed before nucleosynthesis do not take up any of the effective 'baryon budget', and so are a viable candidate for dark matter. Black holes are a form of MACHO (Massive Astrophysical Compact Halo Object) dark matter, like the dim stars and comets from the 30s. Searches for these objects have been done with gravitational microlensing, if a dark matter halo is made totally of MACHOs, then one out of every two million stars should be magnified at any given time [1]. The combined results of about 20 years of observations rules out MACHOs between $10^{-8}M_\odot$ and $1000M_\odot$ [9]. The same concept, and in fact these exact constraints, can be used to restrict the mass of primordial black holes constituting dark matter to between $10^{-16}M_\odot$ and $3 \times 10^{-12}M_\odot$ [3], see Figure 2 for a graph of the constraints. A $10^{-16}M_\odot$ black hole would be about the mass of a small asteroid, about the size of an atom, and you'd expect to find an average of one within the volume of the solar system. If one were to pass through the sun, or any planet, it could be detected through the seismic activity it would produce, a relatively small earthquake [3]. The planned Laser Interferome-

ter Space Antenna (LISA) is expected to be able to detect the background gravitational waves produced by the formation of primordial black holes if indeed they were created in the early universe [10].

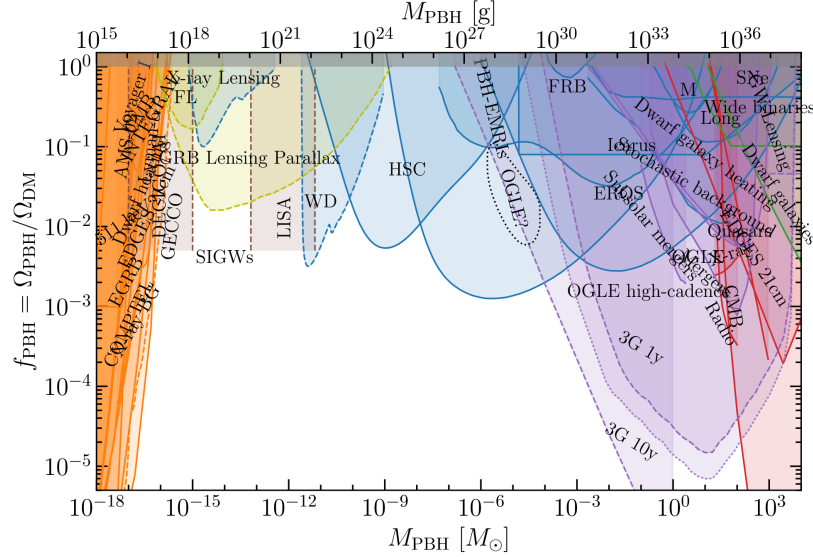


Figure 2: The constraints on the proportion of dark matter that can consist of primordial black holes of a given mass. Important to note is that dotted lines show more debated, less certain constraints. You can see that the only primordial black hole masses that could make up all dark matter are in the range $10^{-16}M_{\odot}$ to $3 \times 10^{-12}M_{\odot}$, discounting the more debated constraints. [11]

4 Direct Detection Experiments

4.1 LUX-ZEPLIN Detector

The LUX-ZEPLIN (LZ) dark matter experiment is the worlds largest dual phase xenon time projection chamber (TPC), a leading version of a widely used technology. This detector involves a large tank of liquid xenon. When a dark matter particle scatters off a nucleus in the liquid, scintillation light is produced, which is picked up by light detectors around the tank. Some detectors stop here, but a TPC applies an electric field across the detector,

causing ionization electrons produced in the event to travel upwards towards a gaseous layer of xenon above the liquid tank. Here, a further electric field is applied, emitting more scintillation photons, creating two signals of different strength: the instant scintillation signal, and the delayed ionization signal. The ratio between these two signals can be used to discriminate between scattering events on electrons, and on nuclei (what the experiment is looking for). [3]

The LZ Experiment is in the Sanford Underground Research Facility in Lead, South Dakota, USA. It uses the TPC setup, detecting scintillation light at the bottom of the detector, electroluminescence at the top of the detector, and the drift time between the detections to find the depth. It uses 7 tonnes of liquid xenon in the active part of the detector, as well as 2 tonnes in a skin around the detector, and an outer detector of 17 tonnes of gadolinium-loaded liquid scintillator. Over 280 days of operation, the most recent results show no excess of events over expected backgrounds, placing the current world leading constraints on the interaction rate between a dark matter particle and a nucleus. The XENONnt experiment in Gran Sasso, Italy, is a similar TPC setup to the LZ experiment, and also has not found an excess of events over the expected background. [12]

4.2 SuperCDMS

The SuperCDMS (Cryogenic Dark Matter Search) is a cryogenic dark matter detector in SNOLAB, Canada. It has 18 Ge detectors and 8 Si detectors: about 30kg of detectors mass, kept at 15 mK. It detects the heat and ionization from the scattering event, allowing it to discriminate between the nuclear and electron recoils. The detector has not yet begun taking data, but a partial test run called CUTE has begun. When completed it will be a world leading dark matter detector for lower mass WIMPS, in the range below 10 GeV, and also perform searches for axions. [13]

4.3 DAMA/LIBRA

DAMA/LIBRA is most famous for it's controversial seasonal variation result. It is a thallium doped sodium iodide NaI(Tl) scintillator crystal detector. When the dark matter particle hits a nucleus in the crystal, the recoil creates light by exciting and de-exciting the atoms in the crystal [14]. Because it is hard to reject false positives in a scintillator only detector, their results come

to gravity, and in galactic centres [3]. An increased neutrino flux in these directions is an indirect detection of dark matter. IceCube is a huge neutrino observatory under the ice in the Antarctic, using the Antarctic ice itself as the detector medium. Neutrinos very rarely interact with matter, so a square kilometer detector like this one is very useful for neutrino physics. When a neutrino interaction happens in the ice, many charged particles are produced, moving in the same direction. These charged particles, due to traveling faster than the speed of light in ice, emit Cherenkov photons which are picked up by 5160 light detectors in 86 boreholes drilled from 1450m to 2450m into the ice, spaced 17m apart, shown in Figure 3. The path of the neutrino can be reconstructed from the tracks produced by the charged particles [17]. No excess in dark matter particles has yet been observed, putting constraints on dark matter models [18], the observatory is being upgraded with more optical modules, which will be able to put unprecedented bounds on dark matter models [19].

5.2 Fermi Gamma-ray Space Telescope

Gamma-rays are another possible particle resulting from the interaction and decay of dark matter. The Fermi gamma-ray telescope is a space telescope capable of detecting gamma rays up to 300 GeV. It looks for an excess of gamma rays coming from the galactic core, or from other nearby galaxies. The Fermi telescope finds a gamma-ray excess with a 40σ significance [20], but there is debate about whether this result can be attributed to dark matter, or results from some other phenomenon such as millisecond pulsars, which also fit the data. It also shows the excess as more flat than spherical, not matching dark matter assumptions, but this is also debated. [20] [21].

5.3 High Energy Stereoscopic System

The High Energy Stereoscopic System (HESS) is a gamma-ray observatory located on the surface of earth. HESS is an array of telescopes in Namibia, made to detect Cherenkov radiation in the atmosphere created by charged particles moving very fast after a gamma-ray event. It only looks at a small portion of the sky, unlike space gamma-ray observatories, but can detect very high energy photons, up to 70TeV [22]. The next generation ground gamma-ray observatory is the Cherenkov Telescope Array Observatory (CTAO), made of 2 large arrays of telescopes, currently under construction. CTAO is

expected to be a huge step for gamma-ray astronomy, with unprecedented sensitivity, resolution, a wider field of view, and an energy range from 20GeV to 300TeV. HESS has been, and hopefully CTAO will be, instrumental in the constraints on high mass dark matter, providing 4 times better limits on 300GeV dark matter than the Fermi space telescope [22] [23].

6 Conclusion

Despite over 50 years of dark matter searches in multiple physics disciplines, there is still little consensus about what dark matter could even theoretically be, and lots of uncertainty in it's true identity. The future, however, of dark matter is bright, ironically. I have mentioned some of the big upcoming experiments motivated entirely or in part by the search for dark matter (LISA, SuperCDMS, CTAO), but the field is very active, and many other projects are being worked on. I didn't mention, for example, collider searches at the LHC, which is currently being upgraded partly for dark matter experimentation [24]. It is hard to tell how close dark matter is to being a solved problem, likely a long way off, but the problem of missing mass in our universe is not going to go away any time soon, and the bounds of the problem are constantly being pushed inwards, without much sign of slowing down; it will be lively and hot area of physics for a very long time.

References

- [1] Gianfranco Bertone and Dan Hooper. History of dark matter. *Reviews of Modern Physics*, 90(4), October 2018.
- [2] S. D. M. White, C. S. Frenk, and M. Davis. Clustering in a neutrino-dominated universe. , 274:L1–L5, November 1983.
- [3] Marco Cirelli, Alessandro Strumia, and Jure Zupan. Dark matter, 2025.
- [4] Douglas Clowe and et al. A direct empirical proof of the existence of dark matter. *The Astrophysical Journal*, 648(2):L109–L113, August 2006.
- [5] Andisheh Mahdavi, Henk Hoekstra, Arif Babul, David D. Balam, and Peter L. Capak. A dark core in abell 520. *The Astrophysical Journal*, 668(2):806–814, October 2007.

- [6] L. Constantin, S. Kraml, and F. Mahmoudi. The lhc has ruled out supersymmetry – really? *Nuclear Physics B*, 1018:117012, September 2025.
- [7] Benoit Famaey, Lorenzo Pizzuti, and Ippocratis D. Saltas. Nature of the missing mass of galaxy clusters in mond: The view from gravitational lensing. *Physical Review D*, 111(12), June 2025.
- [8] Ali Rida Khalifeh and Raul Jimenez. Dwarf galaxies without dark matter: constraints on modified gravity. *Monthly Notices of the Royal Astronomical Society*, 501(1):254–260, November 2020.
- [9] Przemek Mróz and et al. Microlensing optical depth, event rate, and limits on compact objects in dark matter based on 20 yr of ogle observations of the small magellanic cloud. *The Astrophysical Journal Supplement Series*, 280(2):49, September 2025.
- [10] N. Bartolo, V. De Luca, G. Franciolini, M. Peloso, D. Racco, and A. Riotto. Testing primordial black holes as dark matter with lisa. *Physical Review D*, 99(10), May 2019.
- [11] bradley Kavanagh. Github - bradkav/pbhbounds: Data and code for plotting bounds on the abundance of primordial black holes (pbhs)., Nov 2019.
- [12] J. Aalbers and et al. Dark matter search results from 4.2 tonne-years of exposure of the lux-zeplin (lz) experiment. *Physical Review Letters*, 135(1), July 2025.
- [13] Stefan Zatschler. Status and prospects of the SuperCDMS Dark Matter experiment at SNOLAB. *PoS*, TAUP2023:076, 2024.
- [14] R. Bernabei and et al. The dama/libra apparatus. *Nuclear Instruments and Methods in Physics Research Section A: Accelerators, Spectrometers, Detectors and Associated Equipment*, 592(3):297–315, July 2008.
- [15] J. Amaré, S. Cebrián, D. Cintas, I. Coarasa, E. García, M. Martínez, M. A. Oliván, Y. Ortigoza, A. Ortiz de Solórzano, J. Puimedón, A. Salinas, M. L. Sarsa, and P. Villar. Annual modulation results from three-year exposure of anais-112. *Physical Review D*, 103(10), May 2021.

- [16] Govinda Adhikari and et al. Strong constraints from cosine-100 on the dama dark matter results using the same sodium iodide target. *Science Advances*, 7(46), November 2021.
- [17] M.G. Aartsen and et al. The icecube neutrino observatory: instrumentation and online systems. *Journal of Instrumentation*, 12(03):P03012–P03012, March 2017.
- [18] IceCube Collaboration. Search for gev-scale dark matter annihilation in the sun with icecube deepcore. *Physical Review D*, 105(6), March 2022.
- [19] Dan Hooper and Fabrizio Vassallo. Searching for dark matter annihilation in the sun with the icecube upgrade. *Physical Review D*, 112(3), August 2025.
- [20] Moorits Mihkel Muru, Joseph Silk, Noam I. Libeskind, Stefan Gottloeber, and Yehuda Hoffman. Fermi-lat galactic center excess morphology of dark matter in simulations of the milky way galaxy, 2025.
- [21] FERMI Collaboration. The fermi galactic center gev excess and implications for dark matter. *The Astrophysical Journal*, 840(1):43, May 2017.
- [22] HESS Collaboration. Search for gamma-ray line signals from dark matter annihilations in the inner galactic halo from ten years of observations with h.e.s.s. *Physical Review Letters*, 120(20), May 2018.
- [23] Lucia Rinchuso, Emmanuel Moulin, Aion Viana, Christopher Van Eldik, and Johannes Veh. Dark matter gamma-ray line searches toward the galactic center halo with h.e.s.s. i, 2017.
- [24] Sweta Baradia, Sanchari Bhattacharyya, Anindya Datta, Suchandra Dutta, Suvankar Roy Chowdhury, and Subir Sarkar. Search for dark matter in association with a higgs boson at the lhc: A model independent study, 2025.